

PQ4DiT: Time-Aware Polynomial Clipping for Post-Training Quantization of Stable Diffusion 3

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Abstract

Stable Diffusion 3’s (SD3) multi-modal diffusion transformer (MM-DiT) backbone inherits two quantization obstacles from the broader family of diffusion transformers: per-layer activation distributions vary substantially across the denoising schedule, and the presence of salient channels dominates each layer’s dynamic range. We propose a lightweight, time-aware activation clipping scheme to improve on PTQ4DiT’s Channel-wise Saliency Balancing (CSB) and Spearman-correlation-guided Saliency Calibration (SSC) weight transforms. Concretely, each layer’s activation clipping range is modeled as a low-degree polynomial in the noise level σ , and further refined by a per-layer grid search to minimize the reconstruction loss. Our ablations on SD3-medium at W4A8 show that the polynomial clipping scheme yields no substantial improvement over the PTQ4DiT baseline (insignificant differences in CMMD and LPIPS). We interpret this as evidence that CSB/SSC sufficiently compresses the per-channel dynamic-range variation that a static activation scale is competitive on this architecture. As positive secondary contributions we release, to our knowledge, the first public W4A8 PTQ benchmark on SD3 MM-DiT, covering a matrix of pre-quantization transforms, activation quantizers, and weight rounders, together with a reference MLX/Apple Silicon implementation.

1 Background and Context

Diffusion models and their noise level σ . Modern text-to-image diffusion models learn to invert a fixed noising process that maps a clean image $\mathbf{x}_0$ to a sequence of increasingly noisy tensors $\{\mathbf{x}_\sigma\}$ indexed by a continuous noise level $\sigma \in [\sigma_{\min}, \sigma_{\max}]$. A neural network $\epsilon_\theta(\mathbf{x}_\sigma, \sigma, c)$ with parameters θ is trained to predict the added noise at every σ , conditioned on prompt c . At inference, a sample $\mathbf{x}_{\sigma_{\max}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is integrated by a deterministic or stochastic solver down to $\sigma_{\min}$ in tens of network evaluations. The *same* θ is used at every step, but the network’s internal activations depend strongly on σ : at high σ the model processes coarse, high-magnitude signal, while at low σ it refines fine detail at much smaller amplitude. This single-model, many-scales regime is central to the quantization challenges described below.

MM-DiT backbone in Stable Diffusion 3 (SD3).

Early image diffusion models used convolutional U-Nets. Modern systems instead apply a stack of transformer blocks to sequences of spatial patch tokens, yielding the Diffusion Transformer (DiT) family. SD3 [2] extends DiT to a Multi-Modal DiT (MM-DiT): within each block, image and text tokens pass through two separate sub-blocks (an *image transformer block* and a *text transformer block*) whose self-attention paths interact through a joint softmax, and each sub-block’s linear layers are modulated by adaptive layer-norm (adaLN) parameters predicted from the current σ embedding. SD3-medium stacks 24 such blocks, contains $\sim 2.1\text{B}$ parameters, and occupies $\approx 4.17\text{GB}$ in FP16; parameter memory and weight-memory bandwidth therefore dominate the cost of running it across the many forward passes required per generated image.

Uniform integer quantization. Reducing precision below FP16 trades numerical fidelity for memory footprint and bandwidth. A b -bit *symmetric uniform* quantizer with

scale $s > 0$ maps a real tensor $\mathbf{X}$ onto an integer grid of spacing s and decodes back to

$$\hat{\mathbf{X}} = \text{clip}(\text{round}(\mathbf{X}/s), -2^{b-1}, 2^{b-1}-1) \cdot s, \quad (1)$$

so that the representable range $[-s \cdot 2^{b-1}, s \cdot (2^{b-1}-1)]$ is called the *clipping range* of the quantizer. The resulting error has two distinct sources: a *rounding error* on values that fall inside the clipping range (irreducible at a given b , of order $s/2$ per entry), and a *clipping error* on values whose magnitude exceeds the range and are therefore irreversibly saturated. The two errors are coupled through the single knob s : a small s gives fine grid resolution but saturates outliers, a large s preserves extremes but coarsens the grid. An *affine* (asymmetric) variant introduces an integer zero-point $z \in \{0, \dots, 2^b-1\}$, giving $\hat{\mathbf{X}} = s \cdot (\text{clip}(\text{round}(\mathbf{X}/s) + z, 0, 2^b-1) - z)$, which fits distributions that are not centred at zero. The scale s (and, in the affine case, the zero-point z) can be shared at different granularities: one set per tensor (*per-tensor*), per output channel (*per-channel*), or per contiguous group of g input channels (*per-group* with group size g); finer granularity reduces quantization error at a small storage cost.

Weight vs. activation quantization and the diffusion challenge. In a quantized linear layer $\mathbf{Y} = \mathbf{W}\mathbf{X}$, both $\mathbf{W}$ and $\mathbf{X}$ must be represented in low precision. Weight quantization is largely a *static* problem: $\mathbf{W}$ is fixed after training, so its scale (and zero-point, if affine) can be chosen once from the weight tensor itself and refined offline by post-training rounding methods [4, 8, 10]. Activation quantization is *data-dependent*: the scale for $\mathbf{X}$ must be *calibrated* from a representative input set, and any mismatch with the deployment distribution directly inflates error. We denote a pipeline’s precision as $\text{W}b_w\text{A}b_a$; *W4A8* specifies $b_w=4$ -bit weights and $b_a=8$ -bit activations, the target of this work. In diffusion models, activation cali-

bration acquires an additional dimension: the per-layer absolute-maximum (absmax) activation is itself a function of σ and can vary by roughly an order of magnitude across the schedule for a single layer [7, 12], so a single static scale is simultaneously too tight at some σ and too loose at others. This is the core obstacle that motivates the time-aware activation clipping developed in Section 3.

2 Prior Work

2.1 Quantization for Diffusion Models

Quantization for diffusion models is a well-studied field, yet the rapid architectural evolution of generative models, continually demands new methods to keep pace. Diffusion models are expensive to train from scratch, making post-training quantization (PTQ) a more favorable option than quantization-aware training (QAT). Many foundational PTQ techniques from general deep learning have since been adopted by the diffusion community. AdaRound [10] attempts to cast the rounding decision of quantization as a discrete optimization problem solved by minimizing a layer-wise reconstruction loss over calibration data. GPTQ [4] adapts a Hessian framework to large language models, performing column-wise W4 weight quantization with minimal accuracy loss; the method transfers directly to transformer-based diffusion models due to the similarity in architecture.

There has been substantial work to make these quantization techniques ‘time-aware’. Q-Diffusion [7] addressed this problem with timestep-aware calibration and shortcut-splitting for skip-connection activations. TDQ [13] trains a lightweight MLP per quantized layer to predict the quantization scale directly from the timestep embedding, conditioning the clipping range on the current noise level without requiring any modification to the base model weights.

More recently, attention has shifted to Diffusion Transformer (DiT) architectures. PTQ4DiT [14] identified two primary quantization challenges in DiTs: salient channels with extreme magnitudes, and temporal variability in activation distributions across denoising timesteps. TaQ-DiT [9] proposed joint weight-activation block reconstruction to avoid the non-convergence issues of separate reconstruction, and introduced time-variance-aware transformations targeting post-GELU activations, whose distributions shift significantly across denoising steps.

2.2 Limitations of Prior Work

Despite progress in time-aware PTQ, most methods treat timestep-varying activations only implicitly or not at all: calibration is typically run over a sampled mixture of timesteps and the resulting static scales are applied uniformly at inference. Some work, like TDQ, obtains per-timestep scales through explicit MLP training, but this introduces a non-trivial supervised learning stage that requires gradient flow through the quantized model [13].

The majority of existing PTQ work for diffusion models targets U-Net-based architectures [7, 12, 13] that are class-conditioned. SD3’s MM-DiT backbone [2], which

contains a multi-modal backbone, remains largely under-explored. It presents quantization challenges that are not well-studied. Furthermore, weight and activation quantization are typically designed and evaluated independently in the literature. Few pipelines combine state-of-the-art weight PTQ (e.g., GPTQ) with a principled, time-aware activation clipping scheme in a unified framework.

We investigate these gaps with three methodological components and evaluation on SD3 MM-DiT. First, we introduce a *lightweight, time-aware activation clipping* scheme in which each layer’s clipping range is expressed as a low-degree polynomial function of the noise level σ . Second, we combine this scheme with PTQ4DiT’s CSB and SSC, adapted to SD3’s dual-stream image/text architecture, with the aim of absorbing the salient-channel outliers to stabilize the activation distribution. Third, we perform an α -search refinement that minimizes per-layer reconstruction loss on calibration activations. Applied and evaluated on *SD3 MM-DiT*, the resulting empirical study provides, to our knowledge, the first public W4A8 PTQ benchmark on this architecture.

3 Methodology

A. Polynomial Clipping

We express each layer’s clipping range as a low-degree polynomial in the noise level σ . For each quantized layer l , calibration passes record the per-tensor absmax $m_l(\sigma)$ across the denoising schedule. We fit a polynomial by ordinary least squares,

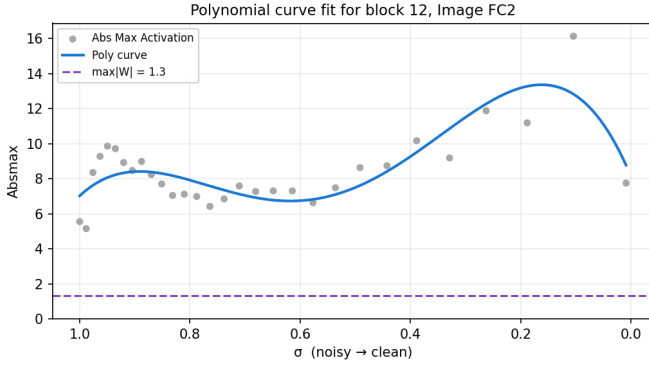
$$\text{poly}_l(\sigma) = \sum_{k=0}^{d_l} c_{l,k} \sigma^k, \quad (2)$$

and use $\text{poly}_l(\sigma)$ as the clipping range of the symmetric uniform 8-bit activation quantizer at inference. The degree is chosen per layer: layers with coefficient of variation $\text{CV}_l < 0.10$ are assigned $d_l = 0$ (static scalar), while more variable layers receive $d_l \in \{2, 3, 4\}$, selected by relative fit-residual reduction and capped at 4 to avoid overfitting.

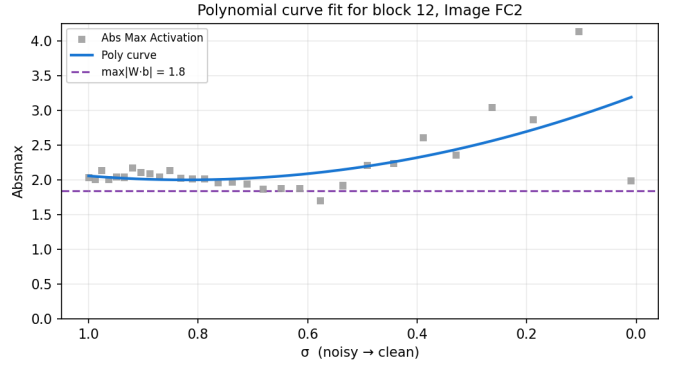
This approach stores only d_l+1 coefficients per layer and is portable across inference step counts: since $\text{poly}_l(\sigma)$ is defined on the continuous σ domain, switching schedulers requires no recalibration.

B. Post-Training Quantization for Diffusion Transformer

Observation: Limitations of Polynomial Clipping on Raw Activations. Applied to raw MM-DiT activations, the polynomial schedule of Section A suffers two pathologies. First, a small number of *salient channels* dominate the per-tensor absmax and inflate the clipping range at the expense of the non-salient majority [14], and their identity drifts across timesteps so no single smooth trajectory captures them. Second, the polynomial fit *smooths* the absmax trajectory, raising its minima and lowering its maxima relative to the observed data (Figure 1(a)), with the



(a) Before CSB/SSC.



(b) After CSB/SSC.

Figure 1: Polynomial fit on the image-stream `fc2` layer at block 12, before (left) and after (right) the PTQ4DiT transforms. Dashed purple lines mark the max-absolute weight entry: $\max_{i,j} |\mathbf{W}_{ij}| = 1.3$ before rescaling, and $\max_{i,j} |(\text{diag}(\mathbf{b})\mathbf{W})_{ij}| = 1.8$ after. Pre-transform, the polynomial smooths the trajectory (absmax range ~ 5 – 16 compressed to ~ 7 – 13) and activation absmax dwarfs the weight scale; after CSB/SSC the absmax range contracts to ~ 2 – 4 , activation and weight scales become comparable, and the polynomial fits tightly.

resulting distortion propagating downstream. We address both with PTQ4DiT’s pre-quantization transforms [14], Channel-wise Saliency Balancing (CSB) and Spearman-correlation-guided Saliency Calibration (SSC), applied *before* the polynomial fit. Because SD3’s MM-DiT routes image and text tokens through separate linear sub-blocks with statistically distinct activations, we apply CSB and SSC independently on each pathway with modality-specific statistics.

CSB and SSC. CSB rewrites each linear layer $\mathbf{Y} = \mathbf{X}\mathbf{W}$ as the exact decomposition $\mathbf{Y} = (\mathbf{X} \oslash \mathbf{b})(\text{diag}(\mathbf{b})\mathbf{W})$, where a per-input-channel balancing vector $\mathbf{b}$ migrates dynamic range from activation to weight channels. Each $\mathbf{b}_j$ is set from the ratio of a schedule-aggregated activation absmax to the weight absmax on channel j , raised to an exponent $\beta \in [0, 1]$ that controls how the rescaling is split; we use the symmetric $\beta = 0.5$ throughout. SSC then specifies how the per-timestep activation absmax is aggregated across the denoising schedule before entering that ratio: steps at which the cross-channel rankings of activation and weight absmax disagree, as measured by a low Spearman correlation ρ_t , are upweighted through a softmax in $-\rho_t$ at temperature τ , so $\mathbf{b}$ is shaped by the hardest timesteps rather than by bulk averages. We use $\tau = 1$, matching the original PTQ4DiT formulation [14]. The full definitions, the clamp bounds on $\mathbf{b}$, and the QKV sharing rule for fused attention projections are deferred to Appendix A.

Polynomial Fit After CSB and SSC. Refitting the polynomial schedule on the CSB/SSC-transformed activations resolves both pathologies (Figure 1(b)): the absmax range contracts $\sim 4\times$, activation and rescaled-weight scales sit within a factor of two, and the polynomial tracks the trajectory tightly. The refitted $\text{poly}_l(\sigma)$ serves as the activation-side input to the per-layer clipping-scale refinement described next.

C. Per-Layer Clipping Scale Refinement via Alpha Search

Observation: Residual Outlier Misfit in Polynomial Clipping. A polynomial fitted on per-layer activation absmax provides a principled time-aware clipping range but is not optimized to minimize quantization error: outlier channels, whether attenuated by the CSB/SSC transforms of Section B or present in raw activations, retain residual influence that can skew the polynomial fit in two opposing ways. *Overclipping* occurs when the polynomial tracks extreme outliers too closely, widening the clipping range and wasting quantization bins on rarely-occupied regions of the activation space. *Underclipping* occurs when the fit is too conservative and truncates activations that carry meaningful signal. Figure 4 illustrates the downstream effect: an image generated without α refinement exhibits blurred facial features, while the same image after α search recovers fine detail.

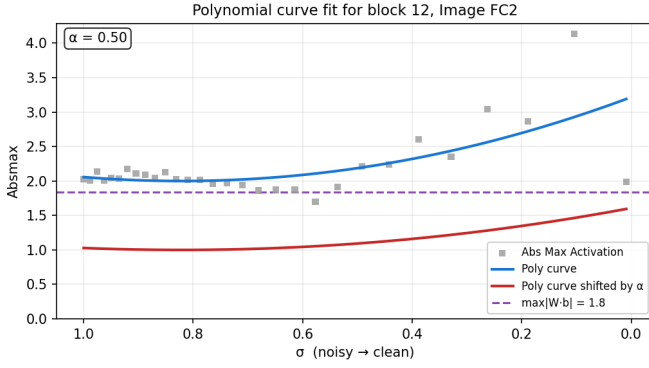
Method: Grid Search over Per-Layer Clipping Scale Multipliers. We introduce a scalar multiplier $\alpha_l \in \mathcal{A}$ that scales the polynomial clipping range of layer l at every noise level, giving an effective clipping range

$$c_l(\sigma) = \alpha_l \cdot \text{poly}_l(\sigma), \quad (3)$$

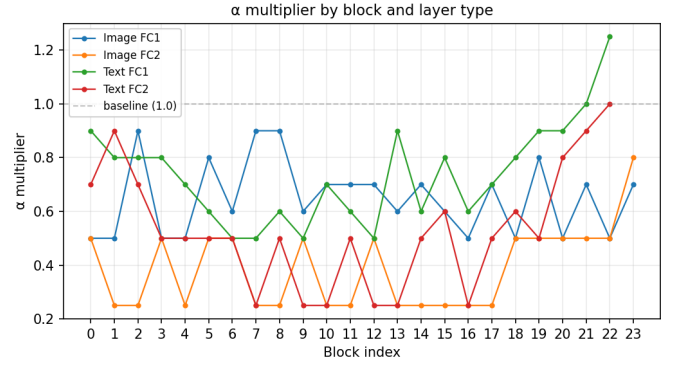
and select the α_l that minimizes the per-layer reconstruction loss over calibration prompts $\mathcal{P}$ and denoising timesteps $\mathcal{T}$:

$$\alpha_l^* = \arg \min_{\alpha \in \mathcal{A}} \sum_{p \in \mathcal{P}, \sigma \in \mathcal{T}} \left\| \hat{\mathbf{X}}_l(\alpha; \sigma) - \mathbf{X}_l(\sigma) \right\|_F^2, \quad (4)$$

where $\mathbf{X}_l(\sigma)$ is the FP16 layer output and $\hat{\mathbf{X}}_l(\alpha; \sigma)$ is the same layer’s output under clipping range $c_l(\sigma)$. $\alpha_l^* > 1$ indicates the polynomial range was too tight (retaining more outlier information outweighs grid coarsening), while $\alpha_l^* < 1$ indicates it was too wide. The full procedure is given in Algorithm 1 in the Appendix. Figure 2a shows a



(a) Polynomial schedule before and after α -search.



(b) Optimal α_i^* by MM-DiT block index.

Figure 2: (a) Polynomial clipping schedule for the image-stream **fc2** layer at block 12 before (blue) and after (red) per-layer α -search, with $\alpha_i^* = 0.5$ tightening the clipping range. (b) Optimal per-layer α_i^* by MM-DiT block index for the image-stream **fc1** and **fc2** MLP layers; most α_i^* fall below 1.0, consistent with absmax-fitted polynomials over-estimating the clipping range, and the optimum varies substantially across blocks within a single layer type.

representative example on the image-stream **fc2** layer at block 12: the absmax-fitted polynomial overestimates the required range, and the search returns $\alpha^* = 0.5$. Because the polynomial is fit on the activation absmax, most α_i^* values fall below 1.0, and the optimum varies considerably across blocks even for layers of the same type (Figure 2b).

4 Experimental Setup

Model and Inference. All experiments use SD3-medium [2] at 512×512 resolution with 30 Euler steps, CFG 4.0, and the rectified-flow σ schedule from 1.0 to 0.001.

Quantization Setup. We target *W4A8*, applied to every `nn.Linear` inside the MM-DiT blocks (embedders, adaLN modulation, and the final projection stay FP16). Weights use an affine per-group grid (group size 32, input-channel axis); activations use a symmetric per-tensor scale, replaced by $\text{poly}_l(\sigma)/127$ for polynomial clipping (Section A) and further scaled by α_i^* for α -search (Section C). All configurations share this grid, so Table 1 is a quality-oriented PTQ ablation, not a runtime benchmark; full backend details are in Appendix D.

Calibration. All data-dependent calibration (PTQ4DiT balancing vectors **b**, polynomial coefficients, per-layer α_i^* , and weight rounding for GPTQ / AdaRound) is computed on a shared set of 100 MS-COCO prompt/seed pairs, sampled once in FP16 under the inference settings above. CSB and SSC are computed independently on the image and text sub-blocks of every MM-DiT block with modality-specific statistics; the remaining calibration hyperparameters match the PTQ4DiT defaults (Appendix D).

Hardware and Software. All calibration, quantization, and inference are implemented in MLX on Apple Silicon (MacBook Pro M1, M4, M5); no CUDA or discrete GPU is used anywhere in the pipeline.

Evaluation Protocol. Benchmarks are computed on an evaluation set of randomly sampled 512 text prompts from

MS-COCO. Using our evaluation prompts, we evaluate the following four metrics:

- **FID** [5]: Fréchet Inception Distance between the quantized-model generations and ground-truth reference images (lower is better).
- **CMMD** [6]: CLIP Maximum Mean Discrepancy, the squared MMD distance computed between CLIP embeddings of the quantized-model generations and the ground-truth reference images (lower is better).
- **LPIPS** [15]: perceptual distance between each quantized image and its FP16 counterpart at the matched seed, reported as mean (std) over the 512 pairs (lower is better).
- **CLIP score** [11]: mean (std) cosine similarity between each generated image and its input prompt in CLIP image-text embedding space (higher is better).

FID is biased and has high-variance at small sample sizes, with reliable estimates typically requiring tens of thousands of samples [6]. Therefore we treat FID as a secondary, corroborating metric and base primary comparisons on CMMD, which was specifically designed to be more sample efficient; LPIPS and CLIP serve as auxiliary perceptual and prompt-alignment indicators respectively.

Pipelines Compared. Each configuration is specified by three independently-chosen components, which lets us isolate the contribution of every component of our pipeline:

- *Pre-quantization transform*: none, or PTQ4DiT (abbreviated **P4D**), which applies CSB and SSC [14].
- *Activation quantizer* (A8): Round-To-Nearest (RTN), polynomial clipping (Poly, Section A), or polynomial clipping followed by per-layer alpha search (Poly + α , Section C).
- *Weight quantizer* (W4): RTN, GPTQ [4], or

Table 1: W4A8 quantization of SD3-medium, evaluated with 30 Euler steps and CFG 4.0 on 512 held-out COCO-caption prompts at 512×512 . FID and CMMD are against ground-truth images; LPIPS pairs each quantized image with the matched-seed FP16 image. LPIPS and CLIP are reported as mean (std). Bold indicates the best quantized result per metric.

Pipeline	FID ↓	CMMD ↓	LPIPS ↓	CLIP ↑
FP16 (reference)	105.87	0.507	–	0.269 (0.032)
<i>Static-calibration baselines</i>				
Vanilla W4A8	250.85	2.696	0.667 (0.075)	0.160 (0.034)
GPTQ	106.53	0.684	0.406 (0.105)	0.265 (0.032)
<i>PTQ4DiT baseline</i>				
P4D [14]	103.80	0.553	0.384 (0.114)	0.266 (0.033)
<i>Polynomial clipping without P4D</i>				
Poly + GPTQ	357.92	3.457	0.729 (0.056)	0.119 (0.024)
Poly + AdaRound	142.64	1.498	0.558 (0.077)	0.238 (0.036)
Poly + MW AdaRound	145.57	1.436	0.565 (0.083)	0.238 (0.036)
Poly + α	109.75	0.858	0.476 (0.097)	0.259 (0.033)
Poly + α + GPTQ	104.83	0.660	0.406 (0.109)	0.266 (0.033)
<i>Polynomial clipping with P4D (ours)</i>				
P4D + Poly	106.32	0.662	0.394 (0.109)	0.265 (0.032)
P4D + Poly + AdaRound	107.12	0.697	0.440 (0.095)	0.268 (0.032)
P4D + Poly + MW AdaRound	107.90	0.730	0.466 (0.095)	0.266 (0.031)
P4D + Poly + α (ours)	104.63	0.556	0.383 (0.116)	0.268 (0.031)

AdaRound [10]. We additionally evaluate a *mean-weighted variant of AdaRound* (“MW AdaRound”) motivated by the polynomial clipping schedule of Section A: MW AdaRound reweights AdaRound’s per-timestep reconstruction loss by the mean absolute derivative of $\text{poly}_l(\sigma)$ over the calibration schedule, concentrating rounding decisions on noise levels where the polynomial clip changes fastest and exerts the most leverage on downstream quantization error. We include it to test whether a time-aware activation clip makes AdaRound more sample-efficient.

Naming convention. We label configurations by their non-default components. RTN is the implicit default for both the A8 activation quantizer and the W4 weight rounder, and is therefore left unstated. When P4D appears in a configuration name, its CSB/SSC pre-quantization transforms are applied first to both activations and weights, and the remaining components then operate on the transformed tensors.

5 Results

Table 1 reports W4A8 results on SD3-medium against the FP16 reference (CMMD 0.507, FID 105.87, CLIP 0.269). Following the protocol above, we discuss the four main findings below.

Vanilla W4A8 fails catastrophically. Plain W4A8 collapses image quality across every metric, with CMMD rising over $5\times$ above the FP16 reference. This confirms that a single per-tensor clipping range cannot simultaneously accommodate the timestep-varying activation magnitudes and the salient-channel outliers of MM-DiT activations, and is the failure mode that motivates polynomial clipping

(Section A) and the PTQ4DiT transforms (Section B) respectively.

PTQ4DiT is a strong baseline. The canonical PTQ4DiT recipe (P4D: CSB/SSC transforms with default RTN rounders on both W4 and A8) recovers CMMD close to the FP16 reference and matches FP16-level CLIP, meaning the quantized model reproduces the FP16 image distribution and responds to prompts comparably. As a separate reference point, W4 GPTQ alone (with default A8 RTN, no P4D) closes most of the W4A8 CMMD gap, showing that a strong weight rounder captures much of what CSB/SSC provides without any activation-side calibration.

Polynomial clipping alone is insufficient; α -search recovers most of its own damage but little beyond. Without CSB/SSC, raw-activation polynomial fits remain corrupted by salient-channel outliers whose identity drifts across timesteps (Section B): every Poly-only row in Table 1 worsens CMMD, and Poly + GPTQ is strictly worse than even the uncalibrated W4A8 baseline on every metric. Adding α -search recovers most of this lost quality, pulling Poly + α + GPTQ back near the PTQ4DiT baseline. But GPTQ alone already reaches comparable CMMD, so α -search here compensates for the degradation polynomial clipping itself introduces rather than extracting a meaningful independent gain.

Adding polynomial clipping on top of PTQ4DiT does not help: the full pipeline only matches the baseline within evaluation noise. Our full pipeline P4D + Poly + α matches the PTQ4DiT baseline across all four metrics within the evaluation noise floor at $n=512$, so we do not claim a quality improvement. The more informative result is the intermediate P4D + Poly row:

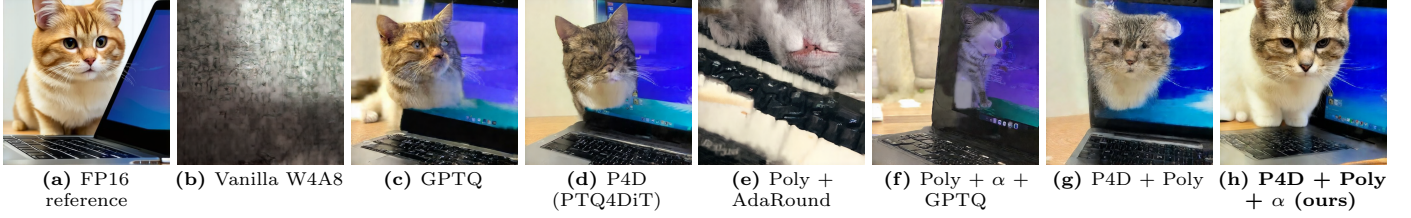


Figure 3: Qualitative comparison across seven W4A8 configurations and the FP16 reference on SD3-medium, generated with 30 Euler steps and CFG 4.0 at matched prompt and seed. Left to right: FP16 reference, vanilla W4A8, GPTQ, the PTQ4DiT baseline (P4D), Poly + AdaRound, Poly + α + GPTQ, P4D + Poly (no α), and our full pipeline P4D + Poly + α . Prompt: “A cat looking like it is using a laptop.” Seed: 305.

introducing polynomial clipping on top of the PTQ4DiT transforms, actively *degrades* CMMD relative to the baseline, and α -search subsequently brings the full pipeline back in line. The α -search step therefore compensates for the degradation polynomial clipping itself introduces rather than extracting additional headroom from time-awareness. Once CSB/SSC is applied, time-aware polynomial activation clipping is redundant on this architecture at this bit budget: CSB/SSC’s per-channel dynamic-range compression already makes a static per-tensor 8-bit activation scale competitive. Within the P4D + Poly family, substituting AdaRound or MW AdaRound for the default RTN weight rounder does not improve over plain RTN either, suggesting marginal returns from a more elaborate weight rounder once CSB/SSC is in place.

Qualitative comparison. Figure 3 shows representative generations from a shared prompt and seed under eight configurations that anchor the quantitative story above, ordered to mirror the progression of Results paragraphs 1–4. The top row tracks paragraphs 1 and 2: the FP16 reference, the catastrophic vanilla W4A8, the GPTQ reference point, and the PTQ4DiT baseline. The bottom row tracks paragraphs 3 and 4: Poly + AdaRound and Poly + α + GPTQ illustrate Poly’s failure without P4D and the subsequent α -search rescue; P4D + Poly (no α) shows that polynomial clipping degrades the PTQ4DiT baseline; and our full pipeline P4D + Poly + α recovers to baseline quality.

6 Key Takeaways

The primary takeaway is a negative result; two secondary positive takeaways follow from the experimental matrix.

6.1 Key Learnings

- **Diffusion quantization is inherently difficult.** Activation distributions shift significantly across denoising timesteps and are prone to extreme outliers.
- **Two core trade-offs govern design choices.**
 - *Quantization:* wider clipping retains outliers at the cost of inefficient bin allocation; tighter clipping improves bin efficiency but discards signal.
 - *Computation:* higher-quality calibration with greater memory requirements improves model performance

but increases inference cost.

- **Polynomial clipping is not a standalone solution.** It relies on PTQ4DiT’s weight stabilization transforms (CSB, SSC) to suppress outliers first, and on alpha search to correct residual per-layer misfit. Each component is necessary for the pipeline to be effective.
- **Quantization methods are architecture-sensitive.** Techniques optimized for U-Net-based diffusion models do not transfer directly to transformer-based architectures such as SD3 MM-DiT, underscoring the need for architecture-aware calibration.

6.2 Conclusion

This work presents the first public W4A8 PTQ benchmark on SD3 MM-DiT, covering 12 quantized configurations across pre-quantization transforms, activation quantizers, and weight rounders. Our primary finding is that time-aware polynomial activation clipping does not improve over PTQ4DiT’s CSB/SSC transforms at W4A8 precision: adding the polynomial schedule degrades CMMD, and while alpha search partially recovers quality, the gain over a naive RTN baseline remains insignificant. This suggests that CSB/SSC already stabilizes per-channel activation dynamics sufficiently to render further time-aware clipping redundant at this bit budget. As a secondary contribution, we release a fully self-contained MLX implementation of the pipeline that runs end-to-end on consumer Apple Silicon without any CUDA dependency, lowering the barrier for future PTQ research on MM-DiT architectures.

7 Future Work

Higher Activation Bitwidth (W8A8). Our experiments focus on W4A8 precision. Evaluating W8A8 would isolate the contribution of weight quantization from activation quantization, providing a cleaner ablation of each component’s impact on image quality and offering a more direct comparison to prior work that targets 8-bit activations.

Generalization to Different Diffusion Architectures. The polynomial clipping schedule was designed and evaluated exclusively on transformer blocks. Extending it to other diffusion architectures such as U-Net-based models (e.g., SD 1.5) would assess whether time-aware polynomial

clipping generalizes across different backbone designs.

Non-Uniform Quantization with Polynomial Clipping. The current pipeline uses a uniform linear quantization grid. Replacing this with non-uniform schemes such as NF4 [1] or logarithmic quantization, which better match heavy-tailed or Gaussian activation distributions, could further reduce reconstruction error within the same bitwidth budget. The polynomial clipping schedule is orthogonal to the choice of grid and could be applied directly on top of such methods.

Replacing Grid Search with Learned Scales. Alpha search over a fixed search space introduces a quantization of the optimal scale itself. Replacing the grid search with a gradient-based approach such as LSQ [3], which treats the clipping scale as a learnable parameter optimized via straight-through estimation, could yield more precise per-layer scales and reduce the number of calibration forward passes required.

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A Full CSB and SSC Formulation

This appendix gives the full definitions of the PTQ4DiT [14] pre-quantization transforms summarized in Section B (“CSB and SSC”), together with the implementation detail required to apply CSB to SD3 MM-DiT’s fused attention projections.

A.1 Channel-wise Saliency Balancing (CSB)

For a linear layer with activation $\mathbf{X} \in \mathbb{R}^{N \times C_{\text{in}}}$ and weight $\mathbf{W} \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}$, CSB introduces a per-channel balancing vector $\mathbf{b} \in \mathbb{R}^{C_{\text{in}}}$ that rewrites the matmul $\mathbf{Y} = \mathbf{X}\mathbf{W}$ as

$$\mathbf{Y} = (\mathbf{X} \oslash \mathbf{b})(\text{diag}(\mathbf{b})\mathbf{W}), \quad (5)$$

where $\oslash$ denotes element-wise division broadcast across the channel axis. The decomposition leaves $\mathbf{Y}$ exactly unchanged but redistributes dynamic range between activation and weight. Writing $s(\mathbf{W}_j) = \max_k |\mathbf{W}_{jk}|$ for the per-input-channel absmx of $\mathbf{W}$, and $\bar{s}(\mathbf{X}_j)$ for a schedule-aggregated per-channel activation absmx (defined by SSC below), CSB sets

$$\mathbf{b}_j = \left(\frac{\bar{s}(\mathbf{X}_j)}{s(\mathbf{W}_j)} \right)^\beta, \quad \text{clamped to } [b_{\min}, b_{\max}], \quad (6)$$

so that channels with disproportionately large activation magnitudes are attenuated, with the excess dynamic range absorbed by the rescaled weight matrix $\tilde{\mathbf{W}} = \text{diag}(\mathbf{b})\mathbf{W}$. The exponent $\beta \in [0, 1]$ sets how the rescaling is split between activation and weight; $\beta = 0.5$ yields a symmetric split, which we use throughout.

A.2 Spearman-correlation-guided Saliency Calibration (SSC)

The balancing quality depends strongly on how the per-timestep per-channel activation absmx $s_t(\mathbf{X}_j) = \max_n |\mathbf{X}_{n,j}^{(t)}|$ is aggregated over the denoising schedule into the $\bar{s}(\mathbf{X}_j)$ used in Eq. 6. A simple average over timesteps weights every step equally and dilutes balancing effort at the steps where it matters most, namely those on which the cross-channel activation-saliency ranking disagrees with the weight-saliency ranking and the unbalanced matmul therefore sees the largest dynamic-range mismatch. SSC identifies these steps via the Spearman rank correlation

$$\rho_t = \text{Spearman}(\{s_t(\mathbf{X}_j)\}_{j=1}^{C_{\text{in}}}, \{s(\mathbf{W}_j)\}_{j=1}^{C_{\text{in}}}) \in [-1, 1], \quad (7)$$

and upweights low- ρ_t timesteps via a temperature-scaled softmax over $-\rho_t$,

$$w_t = \frac{\exp(-\rho_t/\tau)}{\sum_{t' \in \mathcal{T}} \exp(-\rho_{t'}/\tau)}, \quad (8)$$

where $\mathcal{T}$ is the set of calibration timesteps and $\tau > 0$ is a temperature controlling how sharply the weights concentrate on low- ρ_t steps ($\tau \rightarrow 0$ selects only the hardest step, $\tau \rightarrow \infty$ recovers a uniform average). The schedule-aggregated per-channel activation absmx is then

$$\bar{s}(\mathbf{X}_j) = \sum_{t \in \mathcal{T}} w_t s_t(\mathbf{X}_j), \quad (9)$$

which is substituted into Eq. 6 to obtain $\mathbf{b}$. We use $\tau = 1$ throughout, matching the original PTQ4DiT formulation.

A.3 Shared $\mathbf{b}$ for Fused Attention Projections

MM-DiT implements each block’s attention with a single fused linear projection $\mathbf{W}_{\text{QKV}} \in \mathbb{R}^{C_{\text{in}} \times 3C_{\text{head}}}$ that maps an input $\mathbf{X}$ to the concatenation $[\mathbf{Q}; \mathbf{K}; \mathbf{V}]$; an analogous fused projection exists separately on the image and text sub-blocks. Because all three output groups share the same input $\mathbf{X}$, CSB must use a single balancing vector $\mathbf{b}$ per fused projection: a per-group $\mathbf{b}$ would require materializing three different rescaled inputs $\mathbf{X} \oslash \mathbf{b}^{(Q)}$, $\mathbf{X} \oslash \mathbf{b}^{(K)}$, $\mathbf{X} \oslash \mathbf{b}^{(V)}$ and break the fused matmul. We therefore compute the per-input-channel weight absmx over all three output groups jointly,

$$s(\mathbf{W}_{\text{QKV},j}) = \max_{k \in \{1, \dots, 3C_{\text{head}}\}} |(\mathbf{W}_{\text{QKV}})_{j,k}|, \quad (10)$$

and plug this into Eq. 6 with the same schedule-aggregated activation absmx $\bar{s}(\mathbf{X}_j)$ from Eq. 9. The resulting shared $\mathbf{b}$ is applied once to $\mathbf{X}$ and absorbed into $\mathbf{W}_{\text{QKV}}$ as $\text{diag}(\mathbf{b})\mathbf{W}_{\text{QKV}}$, leaving the fused matmul structure intact. Image and text sub-blocks retain independent $\mathbf{b}$ vectors computed from their own modality-specific activation statistics.

B Alpha Search Algorithm

Algorithm 1 Per-Layer Alpha Search

Require: Quantized model (PTQ4DiT + polynomial clipping), calibration prompts $\mathcal{P}$, candidate scales $\mathcal{A}$

Ensure: Per-layer optimal scale α_l^* for all layers l

```

for each layer  $l$  do
  for each prompt  $p \in \mathcal{P}$  do
    Compute FP16 reference output  $\mathbf{y}^{\text{ref}}$ 
     $\alpha_l \leftarrow \text{poly}_l(\sigma)$ 
    for each candidate  $a \in \mathcal{A}$  do
      Fake-quantize input  $\mathbf{X}$  with clipping range  $a \cdot \alpha_l$ 
      Accumulate  $\text{MSE}_a += \|\hat{\mathbf{X}} - \mathbf{X}^{\text{ref}}\|^2$ 
    end for
  end for
   $\alpha_l^* \leftarrow \arg \min_{a \in \mathcal{A}} \text{MSE}_a$ 
end for
return  $\{\alpha_l^*\}$ 

```

C Generated Image Comparison



(a) Pre α search.



(b) Post α search.

Figure 4: Generated image from SD3-medium with 30 Euler steps and CFG 4.0, prompt: “A little girl is laying in bed with a book and a cat is next to her.” Without α -search, the clipping schedule fitted on activation absmx is too loose and fine details (facial features) are lost to coarse quantization; after α -search the per-layer scale is tightened and detail is recovered.

D Experimental Setup Details

Quantization grids. W4 weights use asymmetric uniform quantization with per-group scale and zero-point, groups of size 32 along the input-channel dimension, yielding one (scale, zero) pair per group. A8 activations use the symmetric uniform quantizer $\hat{X} = \text{clip}(\text{round}(X/s), -128, 127) \cdot s$ with a single per-tensor scale s per layer; polynomial clipping (Section A) replaces s with $s_l(\sigma) = \text{poly}_l(\sigma)/127$, and α -search (Section C) further scales it by α_l^* .

Execution backend. RTN-rounded configurations (the PTQ4DiT baseline P4D and our full pipeline P4D + Poly + α) evaluate W4 weights through MLX’s `nn.QuantizedLinear`, which stores them in their 4-bit representation and dequantizes at matmul time. GPTQ, AdaRound, and MW AdaRound round full-precision weights onto the same grid but keep them as *fake-quantized* (quantize-then-dequantize) float tensors, so the matmul runs in FP16 on dequantized weights. Activations are always fake-quantized: the static scale, $\text{poly}_l(\sigma)$, and $\alpha_l^* \text{poly}_l(\sigma)$ are each applied as a quantize-then-dequantize step before a float linear. All comparisons in Table 1 therefore share a single W4A8 grid and read as a quality-oriented PTQ comparison of rounding and calibration choices, not a uniform integer-kernel runtime benchmark.

Calibration hyperparameters. CSB uses balancing exponent $\beta = 0.5$ and clamp range $[b_{\min}, b_{\max}] = [10^{-2}, 10^2]$, with the shared QKV balancing vector computed as described in Appendix A.3. SSC uses temperature $\tau = 1$.